Strategic Integration of Computational Bioinformatics and Machine Learning in Phytochemical Discovery for Chronic Wound Therapeutics

The global landscape of cutaneous wound management is undergoing a paradigm shift, transitioning from traditional palliative care to precision-driven regenerative medicine. Chronic wounds, defined by their failure to progress through the physiological stages of repair within a standard three-month timeframe, affect approximately one billion individuals worldwide.[1] These conditions, which include diabetic foot ulcers (DFUs), venous leg ulcers (VLUs), and pressure injuries, impose staggering economic costs on healthcare systems and diminish the quality of life for aging and diabetic populations.[2] For a pharmaceutical graduate seeking to pivot into computational drug discovery, the intersection of ethnopharmacology and bioinformatics provides a unique vantage point to identify novel bioactive molecules from the vast chemical space of phytochemicals.[4] By utilizing a high-fidelity computational pipeline—incorporating Avogadro for ligand optimization, UCSF Chimera and AutoDock Vina for molecular docking, and BIOVIA for workflow automation—researchers can simulate complex biological interactions with atomistic precision.[6] This strategic report outlines an exhaustive framework for an independent research project, incorporating advanced artificial intelligence (AI) and machine learning (ML) methodologies to ensure the work resonates with the rigorous standards of premier research institutions in the United States, United Kingdom, Canada, Australia, and specifically the Scandinavian innovation hubs of Sweden and Norway.[9]

Molecular Pathophysiology of Chronic Wounds and Therapeutic Targeting

Understanding the molecular barriers to healing is the prerequisite for any computational screening endeavor. Normal wound healing is a highly orchestrated process involving four overlapping phases: hemostasis, inflammation, proliferation, and remodeling.[12] In chronic wounds, this orchestration is disrupted by deleterious feedback loops that favor a persistent hyper-inflammatory state.[13]

Disruption of the Coagulation and Inflammatory Cascades

The initial response to vascular injury involves platelet activation and the conversion of fibrinogen to fibrin by thrombin, forming a provisional matrix.[14] In chronic wounds, this fibrin mesh often becomes a site of stagnation rather than a scaffold for migration. Platelets normally

release growth factors such as Transforming Growth Factor-beta (TGF-β), Platelet-Derived Growth Factor (PDGF), and Vascular Endothelial Growth Factor (VEGF) to initiate repair.[3] However, the chronic wound environment is characterized by an overabundance of pro-inflammatory cytokines (IL-1β, TNF-α, and IL-6) and reactive oxygen species (ROS), which degrade these essential signaling molecules before they can enact their regenerative functions.[3]

Matrix Metalloproteinases (MMPs) as Primary Computational Targets

A central feature of non-healing wounds is the dysregulation of Matrix Metalloproteinases (MMPs), a family of zinc-dependent endopeptidases responsible for extracellular matrix (ECM) remodeling.[16] While acute wounds require balanced MMP activity for cell migration and tissue turnover, chronic wounds exhibit an pathological excess of specific MMPs, notably MMP-8 (collagenase-2) and MMP-9 (gelatinase B).[16]

MMP Class	Key Member	Role in Chronic Wounds	Potential Phytochemical Modulation
Collagenases	MMP-8	Highly elevated in DFUs; degrades collagen type I.[17]	Tannic acid and curcumin have shown inhibitory potential.[19]
Gelatinases	MMP-9	Key biomarker for non-healing; degrades the basement membrane.[16]	Flavonoids (e.g., quercetin) inhibit transcription via NF-κB pathways.[5]
Stromelysins	MMP-10	Impaired in chronic wounds; essential for keratinocyte migration.[1]	Seeking enhancers or stabilizers is a novel research direction.[1]
Matrilysins	MMP-7	Involved in mucosal repair; often dysregulated in chronic	Polysaccharides and alkaloids may modulate expression.[16]

		inflammation.[18]	

Recent transcriptomic profiling has identified MMP-10 as a particularly intriguing target.[1] In acute wounds, MMP-10 is expressed at the leading edge of migrating keratinocytes to facilitate re-epithelialization.[18] In chronic states, its expression is often diminished, leading to a failure of wound closure.[1] A computational project focusing on identifying dual-function phytochemicals—those that inhibit the pathological overactivity of MMP-9 while potentially stabilizing the pro-migratory function of MMP-10 or its upstream regulators like Fibroblast Growth Factor 7 (FGF7)—would represent a high-impact, current research trend.[1]

Computational Workflow: From Phytochemical Library to Lead Optimization

For a pharmacy graduate, the ability to execute a rigorous, reproducible computational pipeline is the ultimate demonstration of technical competency. The proposed workflow integrates the tools provided (Avogadro, Chimera, AutoDock Vina, BIOVIA) with free web-based platforms to create an industry-standard drug discovery funnel.[6]

Ligand Selection and Geometrical Optimization

The chemical space for the project should be sourced from specialized natural product databases. **Phytochemdb** provides a manually curated collection of over 8,000 phytochemicals from 525 medicinal plants, including detailed SMILES strings and 3D structures.[4] Other essential repositories include **COCONUT** (400k+ natural products) and **ZINC15 Natural Products**.[24]

The initial step in Avogadro involves building or importing these ligands and performing energy minimization.[8] This process ensures that the molecules are in their most stable, bioactive conformation before docking. Geometry optimization typically utilizes the Universal Force Field (UFF) or Merck Molecular Force Field (MMFF94).[8] The researcher must calculate the molecular descriptors (LogP, molecular weight, H-bond donors/acceptors) to ensure the library complies with drug-likeness criteria such as Lipinski's Rule of Five or the Ghose filter.[25]

Target Preparation and Binding Site Analysis in UCSF Chimera

Target proteins, such as the catalytic domain of MMP-9 (PDB ID: 1L6J) or the FGF7 receptor, are retrieved from the Protein Data Bank (PDB).[7] UCSF Chimera's "Dock Prep" tool facilitates the necessary modifications:

1. **Removal of Solvents and Non-complexed Ions:** Eliminating water molecules (HOH) that might interfere with ligand binding, unless they are known to be structural.[8]
2. **Protonation:** Adding hydrogen atoms to the protein structure based on the predicted

pKa of amino acid side chains at physiological pH.[8]

3. **Charge Assignment:** Utilizing the Gasteiger atom charge method to define the electrostatic potential of the receptor surface.[28]
4. **Inhibitor Delineation:** Identifying any co-crystallized inhibitor to define the grid box coordinates (center and size) for the docking search space.[28]

High-Throughput Docking with AutoDock Vina

AutoDock Vina utilizes a Lamarckian Genetic Algorithm and an empirical scoring function to predict the binding affinity of ligands.[7] The binding energy (ΔG) is calculated using the following LaTeX notation to represent the total energy of the system:

$$\Delta G_{binding} = W_{vdw} \sum_{i,j} \left(\frac{A_{ij}}{r_{ij}^{12}} - \frac{B_{ij}}{r_{ij}^{6}} \right) + W_{hbond} \sum_{i,j} E(t) \left(\frac{C_{ij}}{r_{ij}^{12}} - \frac{D_{ij}}{r_{ij}^{10}} \right) + W_{elec} \sum_{i,j}$$

where W represents weighting factors for van der Waals, hydrogen bonding, electrostatic, and solvation energies.[28] A high-quality project should use an "exhaustiveness" setting of 16 to 32 to ensure the global minimum energy pose is identified.[8]

Workflow Automation with BIOVIA Pipeline Pilot

To boost credibility in corporate and advanced academic environments, the researcher should demonstrate the use of **BIOVIA Pipeline Pilot**.[6] This graphical authoring application allows for the construction of automated protocols that can:

- Standardize large batches of phytochemical SMILES strings.[6]
- Integrate docking scores from Vina with post-processing filters.[6]
- Analyze protein-ligand interactions (e.g., hydrogen bonding, hydrophobic contacts) to rank compounds not just by score, but by mechanistic relevance.[6]
- Deploy Bayesian or Random Forest models to predict the probability of success in subsequent wet-lab experiments.[6]

Advanced Integration of AI and Machine Learning

Incorporating AI/ML is no longer optional for high-impact bioinformatics research.[32] These technologies provide the predictive power to shorten the gap between target identification and candidate optimization.[32]

Protein Structure Prediction with AlphaFold 3

For targets where experimental structures are unresolved (e.g., novel wound-healing receptors or specific isoforms of collagenases), **AlphaFold 3** from Google DeepMind is the gold standard.[33] It provides near-atomic accuracy for protein structure prediction and, more importantly, allows for the modeling of protein-protein interactions and ligand docking with improved accuracy over traditional homology modeling.[33] This demonstrates a keen attention to current trends (2024-2025 releases).[33]

Generative AI for Molecular Design

Rather than just screening existing phytochemicals, the researcher can use generative AI to design "bio-isosteres"—molecules that mimic the activity of a phytochemical but with better drug-like properties.[37] Startups like **Insilico Medicine** and **Exscientia** utilize Generative Adversarial Networks (GANs) and Reinforcement Learning (RL) for this purpose.[34] For an independent project, tools like **DeepChem** (an open-source library) can be used to implement these models.[24]

Explainable AI (XAI) and Network Pharmacology

To satisfy the need for mechanistic transparency, **Explainable AI (XAI)** should be applied to the screening results.[39] XAI helps researchers identify *why* a model predicts a molecule will be active—whether it's due to a specific hydroxyl group or a particular aromatic scaffold—thereby increasing trust and biological relevance.[39] Furthermore, **Network Pharmacology** (using tools like **SwissTargetPrediction** and **STITCH**) allows the researcher to build a "drug-target-pathway" network to identify phytochemicals that act on multiple nodes simultaneously, which is crucial for addressing the complex etiology of chronic wounds.[15]

AI/ML Application	Recommended Tool/Platform	Value Addition
Structural Modeling	AlphaFold 3.[33]	High-accuracy models for non-crystallized targets.[35]
Property Prediction	SwissADME & ProTox-II.[26]	Rapid filtering of toxic or poorly absorbed compounds.[26]
Bioactivity Classification	Random Forest (via Pipeline Pilot).[6]	Predicts IC50 or binding affinity based on molecular fingerprints.[30]
Target Identification	SwissTargetPrediction.[42]	Uncovers off-target effects

		and polypharmacological potential.[42]

Strategic Geographic Alignment: Scandinavia, the UK, and the USA

To grab the attention of international bodies, the research must be contextualized within the strategic priorities of specific regions.[9]

Sweden: Hagastaden and Karolinska Institutet

Sweden is a global hub for healthcare innovation, particularly in the Stockholm-Uppsala region.[9] The **Hagastaden** innovation district is home to the **Karolinska Institutet (KI)**, which hosts the world-renowned **Xu Landén Group**.[47]

- **Current Research Trend:** The Xu Landén Group focuses on the role of regulatory RNAs (miRNA, lncRNA, circRNA) and single-cell transcriptomics in human wound healing.[47]
- **Strategy:** Positioning the phytochemical project as a screening tool to find ligands that modulate the expression of non-coding RNAs or target specific immune-cell subtypes (identified via the KI "Wound Atlas") would be highly compelling for Swedish researchers.[47]

Norway: Blue Biotechnology and Oslo Science City

Norway is leveraging its marine resources to solve global health challenges through bioprospecting.[45]

- **Current Research Trend: MabCent** in Tromsø and **Regenics AS** in Oslo focus on discovering bioactive molecules from Arctic marine organisms for chronic wound therapeutics.[53]
- **Strategy:** Expanding the "phytochemical" interest to include "marine natural products" (using databases like CMNPD) aligns perfectly with the Norwegian goal of "sustainable, knowledge-based blue-green bio-innovation".[24] Furthermore, **Oslo Science City** is currently undergoing massive expansion (The Loop) to become an epicenter for health and life sciences by 2026.[56]

USA, UK, and Australia: Large-scale Computational Integration

In the USA and UK, the emphasis is on the integration of large-scale clinical data with AI-driven drug discovery.[34]

- **Strategy:** Utilizing **PubChem** (USA) and **ChEMBL** (UK/Europe) for bioassay data and adhering to **FAIR (Findable, Accessible, Interoperable, Reusable)** data principles demonstrates the researcher's readiness to work within high-level international consortiums.[59]

Professional Portfolio Development and Credibility Enhancement

As a pharmacy graduate waiting for internship placement, the GitHub portfolio serves as a "digital lab bench" that proves technical capability to international employers.[62]

Anatomical Structure of a Standout Bioinformatics Project

The repository should be modular and follow the best practices of open science.[31]

1. **README.md:** This must include a clear scientific rationale, a flowchart of the computational pipeline, and a summary of the most promising phytochemical hits.[31]
2. **Environment Files:** Inclusion of requirements.txt or environment.yml to allow others to reproduce the analysis pipeline using Python or R.[62]
3. **Data Curation:** A data/ folder containing the filtered phytochemical libraries and the PDBQT files of the target proteins.[62]
4. **Analysis Notebooks:** Jupyter Notebooks labeled by phase (e.g., Part_1_Data_Acquisition.ipynb, Part_2_Docking_Results.ipynb) with extensive comments explaining the biological relevance of each step.[31]
5. **Visualization:** High-resolution renders from ChimeraX or PyMOL showing the docked ligand in the active site, highlighting specific hydrogen bonds or hydrophobic interactions with catalytic residues like Zn^{2+} in the MMP active site.[8]

Boosting Credibility through Open Science and Documentation

To secure a DOI for the project, the repository should be linked to **Zenodo** or **Figshare**.[68] This allows the work to be cited in a formal CV or resume.[68] Furthermore, contributing small pull requests to established open-source projects like **Biopython** or **DeepChem** acts as an immediate badge of credibility, showing that the researcher can collaborate at an international level.[62]

Portfolio Element	Best Practice	Purpose
Code Versioning	Use Git/GitHub with frequent, meaningful commits.[62]	Demonstrates consistency and technical growth.[62]
Documentation	Write detailed "vignettes" for each analytical step.[64]	Proves the ability to communicate complex science.[64]

Reproducibility	Use containerization (Docker) or Conda environments.[64]	Ensures the pipeline works on any international server.[74]
Open Access	Link repository to Zenodo for a persistent DOI.[68]	Makes the independent research "citable".[68]

Conclusion and Strategic Roadmap

The transition from a clinical pharmacy background to computational bioinformatics requires a strategic embrace of both the biological complexity of chronic wounds and the technical rigor of modern drug discovery tools. By focusing on phytochemical inhibitors of pathological MMP-9 while exploring the novel stabilizing potential of MMP-10 regulators, the researcher addresses a critical, unmet need in global health. The integration of AlphaFold 3 for structural modeling and BIOVIA for workflow automation places the project at the technological forefront.

Strategically, aligning the research with the transcriptomic and regulatory RNA interests of Swedish institutions or the marine bioprospecting priorities of Norwegian clusters provides a clear entry point into the Scandinavian innovation ecosystem. When supported by a professionally curated GitHub portfolio that adheres to FAIR principles and features citable DOIs, this independent research project becomes a powerful vehicle for securing international internships, Ph.D. positions, or roles in the burgeoning AI-driven pharmaceutical industry. The future of wound care lies in the digital deconvolution of nature's chemical library, and a pharmacy graduate equipped with these computational skills is uniquely positioned to lead this transformation.

Works cited

1. Breakthrough Study Highlights Key Molecular Targets to Enhance Wound Healing, accessed March 27, 2026, https://provaeducation.com/news/key-molecular-targets-for-wound-healing-identified/2468313/
2. Immunomodulatory Mechanisms of Chronic Wound Healing: Translational and Clinical Relevance - PMC, accessed March 27, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC12538007/
3. Chronic Wound Healing: Research Advances from Pathological ..., accessed March 27, 2026, https://www.mdpi.com/1420-3049/31/6/1024
4. phytochemdb: a platform for virtual screening and computer-aided drug designing | Database | Oxford Academic, accessed March 27, 2026, https://academic.oup.com/database/article/doi/10.1093/database/baac002/6535291
5. Research Hotspots in Natural Products for Wound Healing: A Bibliometric Analysis and Literature Review - PMC, accessed March 27, 2026,

https://pmc.ncbi.nlm.nih.gov/articles/PMC12380089/
6. Enhancing Drug Discovery with BIOVIA Pipeline Pilot - Astrix, accessed March 27, 2026, https://astrixinc.com/blog/enhancing-drug-discovery-with-biovia-pipeline-pilot/
7. Molecular Docking Using Chimera and Autodock Vina Software for Nonbioinformaticians, accessed March 27, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC11168238/
8. AutoDock Vina Tutorial: Protein-Ligand Docking for Beginners 2026, accessed March 27, 2026, https://www.genomatics.net/guides/autodock-vina-tutorial-beginners
9. Sweden's Life Science Ecosystem, accessed March 27, 2026, https://www.business-sweden.com/insights/articles/swedens-life-science-ecosystem/
10. About us | SSCI - Stockholm Science City, accessed March 27, 2026, https://ssci.se/en/about-us
11. Tale på Livsvitenskapskonferansen 2026 - regjeringen.no, accessed March 27, 2026, https://www.regjeringen.no/no/aktuelt/tale-pa-livsvitenskapskonferansen-2026/id3150380/
12. (PDF) Progress in Wound Healing Products based on Natural Compounds, Stem Cells and MicroRNA-based Biopolymers in the European, USA, and Asian Markets: Opportunities, Barriers, and Regulatory Issues - ResearchGate, accessed March 27, 2026, https://www.researchgate.net/publication/380483052_Progress_in_Wound_Healing_Products_based_on_Natural_Compounds_Stem_Cells_and_MicroRNA-based_Biopolymers_in_the_European_USA_and_Asian_Markets_Opportunities_Barriers_and_Regulatory_Issues
13. Rethinking Chronic Wound Treatment: Unlocking the Potential of ..., accessed March 27, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC12883342/
14. Wound Healing: Molecular Mechanisms, Antimicrobial Peptides, and Emerging Technologies in Regenerative Medicine - MDPI, accessed March 27, 2026, https://www.mdpi.com/1424-8247/18/10/1525
15. Network Pharmacology as a Tool to Investigate the Antioxidant and Anti-Inflammatory Potential of Plant Secondary Metabolites—A Review and Perspectives - MDPI, accessed March 27, 2026, https://www.mdpi.com/1422-0067/26/14/6678
16. Targeting matrix metalloproteases in diabetic wound healing - PMC - NIH, accessed March 27, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC9981633/
17. The specific role of extracellular matrix metalloproteinases in the pathology and therapy of hard-to-heal wounds - Frontiers Publishing Partnerships, accessed March 27, 2026, https://www.frontierspartnerships.org/journals/neuroengineering/articles/10.18388/abp.2020_6934/pdf
18. Metalloproteinases and Wound Healing - PMC, accessed March 27, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC4397992/

19. Phytochemicals in Wound Healing - PMC - NIH, accessed March 27, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC4827319/
20. The Role of Phytochemicals in the Inflammatory Phase of Wound Healing - ResearchGate, accessed March 27, 2026, https://www.researchgate.net/publication/317052347_The_Role_of_Phytochemicals_in_the_Inflammatory_Phase_of_Wound_Healing
21. Matrix metalloproteinases in impaired wound healing - Dove Medical Press, accessed March 27, 2026, https://www.dovepress.com/matrix-metalloproteinases-in-impaired-wound-healing-peer-reviewed-fulltext-article-MNM
22. Computational Approaches to Molecular Docking and Protein Modeling in Drug Discovery, accessed March 27, 2026, https://jddtonline.info/index.php/jddt/article/view/7212/6852
23. phytochemdb - Bio.tools, accessed March 27, 2026, https://bio.tools/phytochemdb
24. yboulaamane/awesome-drug-discovery - GitHub, accessed March 27, 2026, https://github.com/yboulaamane/awesome-drug-discovery
25. Virtual Screening of Phytochemicals From Medicinal Plants as Promising PDE5 Inhibitors Against Erectile Dysfunction - PMC, accessed March 27, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC12862184/
26. ADME/ADMET Assessment Using SwissADME, ProTox-II, admet SAR, and Swiss Target Prediction: A Comprehensive Guide - Pars Silico, accessed March 27, 2026, https://parssilico.com/blogs/109-adme-admet-assessment
27. Swiss ADME Predictions of Pharmacokinetics and Drug-Likeness Properties of Chemical Constituents Present in Embelia Ribes - IJPPR, accessed March 27, 2026, https://ijppr.humanjournals.com/wp-content/uploads/2025/01/10.Miss_.-Tasmeem-T.-Shaikh-Dr.-Rahul-S.-Adnaik-Miss.-Neha-B.-Patil.pdf
28. Molecular Docking Workflow with AutoDock Vina and ChimeraX - In Silico Design, accessed March 27, 2026, https://www.insilicodesign.com/en/post/molecular-docking-workflow-with-autodock-vina-and-chimerax
29. Molecular Docking Using Chimera and Autodock Vina Software for Nonbioinformaticians, accessed March 27, 2026, https://bioinform.jmir.org/2020/1/e14232
30. Qsarna: An Online Tool for Smart Chemical Space Navigation in Drug Design, accessed March 27, 2026, https://pubs.acs.org/doi/10.1021/acs.jcim.5c00720
31. shashwat0105/Bioinformatics-Drug-Discovery ... - GitHub, accessed March 27, 2026, https://github.com/shashwat0105/Bioinformatics-Drug-Discovery
32. 10+ Scientific AI Tools Every Scientist Should Know in 2025/26 | Sapio Sciences, accessed March 27, 2026, https://www.sapiosciences.com/blog/10-scientific-ai-tools-every-scientist-should-know-in-2025-26/
33. High-Demand AI/ML Tools in Biology You Need to Know in 2025 - BioTecNika, accessed March 27, 2026,

https://www.biotecnika.org/2025/05/high-demand-ai-ml-tools-in-biology-you-need-to-know-in-2025/
34. Top 7 AI Tools Transforming Drug Discovery and Biotech R&D - Maurizio Chiriva Internati, accessed March 27, 2026, https://mauriziochirivainternati.com/top-7-ai-tools-transforming-drug-discovery-and-biotech-rd/
35. 10 Useful AI Tools for Life Scientists - Hello Bio, accessed March 27, 2026, https://hellobio.com/blog/10-useful-ai-tools-for-life-scientists.html
36. AI Tools Every Biomedical Researcher Should Know in 2025, accessed March 27, 2026, https://irrespub.com/ai-tools-biomedical-research-2025/
37. Top 10 AI Applications in Biotechnology (2025+) | StartUs Insights, accessed March 27, 2026, https://www.startus-insights.com/innovators-guide/ai-in-biotechnology/
38. Top 10 AI Drug Discovery Platforms in 2026: Features, Pros, Cons & Comparison, accessed March 27, 2026, https://www.devopsschool.com/blog/top-10-ai-drug-discovery-platforms-in-2025-features-pros-cons-comparison/
39. Explainable Artificial Intelligence for Drug Discovery and Development - A Comprehensive Survey - arXiv, accessed March 27, 2026, https://arxiv.org/html/2309.12177v1
40. Explainable Artificial Intelligence: A Perspective on Drug Discovery - MDPI, accessed March 27, 2026, https://www.mdpi.com/1999-4923/17/9/1119
41. (PDF) Explainable Artificial Intelligence for Drug Discovery and Development - A Comprehensive Survey - ResearchGate, accessed March 27, 2026, https://www.researchgate.net/publication/378716205_Explainable_Artificial_Intelligence_for_Drug_Discovery_and_Development_-_A_Comprehensive_Survey
42. SwissTargetPrediction - Bio.tools, accessed March 27, 2026, https://bio.tools/SwissTargetPrediction
43. Network Pharmacology: A Systems-Based Paradigm-DrivenMulti-Target Drug Discovery &Development | European Journal of Cardiovascular Medicine, accessed March 27, 2026, https://healthcare-bulletin.co.uk/article/network-pharmacology-a-systems-based-paradigm-drivenmulti-target-drug-discovery-development-3872/
44. ProTox-II: a webserver for the prediction of toxicity of chemicals - PMC, accessed March 27, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC6031011/
45. BioTech North, accessed March 27, 2026, https://www.biotechnorth.no/
46. A concerted effort for long-term excellence, competitiveness and increased patient benefit, accessed March 27, 2026, https://www.government.se/information-material/2025/02/a-concerted-effort-for-excellence-long-term-competitiveness-and-increased-patient-benefit/
47. Skin wound healing – Research group Ning Xu Landén | Karolinska Institutet, accessed March 27, 2026, https://ki.se/en/research/research-areas-centres-and-networks/research-groups/skin-wound-healing-research-group-ning-xu-landen
48. Hagastaden | SSCI - Stockholm Science City, accessed March 27, 2026,

https://ssci.se/en/hagastaden
49. Postdoctoral studies in Computation Biology: Molecular Wound Healing (scholarship), accessed March 27, 2026, https://ki.varbi.com/se/what:job/jobID:917325/
50. Postdoctoral studies in Computation Biology: Molecular Wound Healing (scholarship), accessed March 27, 2026, https://ki.varbi.com/ro/what:job/jobID:917325/
51. Scholarship: Postdoctoral Researcher in Computational Biology for Molecular Wound Healing at Karolinska Institutet | ApplyKite, accessed March 27, 2026, https://www.applykite.com/scholarships/postdoctoral-researcher-in-computational-biology-for-molecular-wound-healing-at-karolinska-institutet-ztx309mwwb
52. Marine Bioprospecting - UiT The Arctic University of Norway, accessed March 27, 2026, https://en.uit.no/go/target/348358
53. Regenics – Innovative wound care technology, accessed March 27, 2026, https://regenics.no/
54. MabCent: Arctic marine bioprospecting in Norway - PMC, accessed March 27, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC3777186/
55. MabCent: Arctic marine bioprospecting in Norway - Norwegian Research Information Repository, accessed March 27, 2026, https://nva.sikt.no/registration/0198cc52c9e0-dddb174f-f73e-42d1-b492-7a073fc6391a
56. Norway Life Science 2026: Building the Future - NMBU, accessed March 27, 2026, https://www.nmbu.no/en/research/events/norway-life-science-2026-building-future
57. Oslo Science City - Bjarke Ingels Group, accessed March 27, 2026, https://big.dk/projects/oslo-science-city-4197
58. Health and life science | Invest in Norway, accessed March 27, 2026, https://businessnorway.com/invest-in-norway/industries/health-and-life-science
59. Full article: Exploiting PubChem and other public databases for virtual screening in 2025: what are the latest trends? - Taylor & Francis, accessed March 27, 2026, https://www.tandfonline.com/doi/full/10.1080/17460441.2025.2558161
60. Artificial Intelligence to Guide Repurposing of Drugs - PMC - NIH, accessed March 27, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC12854511/
61. FAIR Principles, accessed March 27, 2026, https://www.go-fair.org/fair-principles/
62. How to Build a GitHub Portfolio in Bioinformatics: A Step-by-Step Guide for Beginners, accessed March 27, 2026, https://www.biotecnika.org/2025/09/how-to-build-a-github-portfolio-in-bioinformatics-a-step-by-step-guide-for-beginners/
63. How to Build a Bioinformatics Portfolio - Bversity, accessed March 27, 2026, https://bversity.io/blog/how-to-build-a-bioinformatics-portfolio
64. Portfolio Projects That Get You Hired for Biotech Jobs (With Real GitHub Examples), accessed March 27, 2026, https://biotechnologyjobs.co.uk/career-advice/portfolio-projects-that-get-you-hired-for-biotech-jobs-with-real-github-examples-
65. othneildrew/Best-README-Template: An awesome README template to

jumpstart your projects! - GitHub, accessed March 27, 2026, https://github.com/othneildrew/Best-README-Template
66. aws-samples/drug-discovery-workflows - GitHub, accessed March 27, 2026, https://github.com/aws-samples/drug-discovery-workflows
67. drug-discovery · GitHub Topics, accessed March 27, 2026, https://github.com/topics/drug-discovery?l=html
68. Code / Software - FAIR guidance, accessed March 27, 2026, https://sites.google.com/sheffield.ac.uk/fair-guidance/your-data-typecode/code-software
69. Generalist Repository Comparison Chart - Zenodo, accessed March 27, 2026, https://zenodo.org/records/17315963/files/Generalist%20Repository%20Comparison%20Chart_version%204.0.pdf?download=1
70. General Repository Comparison - FAIRsharing, accessed March 27, 2026, https://fairsharing.org/collection/GeneralRepositoryComparison
71. Bioinformatics repository examples with good practices of using GitHub. - Figshare, accessed March 27, 2026, https://plos.figshare.com/articles/dataset/Bioinformatics_repository_examples_with_good_practices_of_using_GitHub_/4013877
72. How much do github projects help with job hunting? : r/bioinformatics - Reddit, accessed March 27, 2026, https://www.reddit.com/r/bioinformatics/comments/1iq75fi/how_much_do_github_projects_help_with_job_hunting/
73. wguesdon/Portfolio: A collection of personal data science and computational biology projects. - GitHub, accessed March 27, 2026, https://github.com/wguesdon/Portfolio
74. minimal example implementations for bioinformatics workflow managers - GitHub, accessed March 27, 2026, https://github.com/GoekeLab/bioinformatics-workflows